

Write a Prolog Program to Sum the Integers from 1 to n

Aim

To develop a Prolog program that calculates the sum of integers from **1 to n** using recursion.

Objective

- To understand recursive programming in Prolog.
- To implement arithmetic operations using recursion.
- To calculate the sum of natural numbers.
- To learn the concept of recursive predicates.

Theory

Prolog (Programming in Logic) is a logic programming language used in Artificial Intelligence. It solves problems using facts, rules, and recursion. Recursive predicates repeatedly call themselves until a base condition is reached. In this experiment, recursion is used to calculate the sum of integers from **1 to n**. The base case occurs when **n = 0**, and the recursive case computes the sum by adding the current number to the sum of the previous numbers.

Algorithm

Step 1: Define the base case where the sum of 0 is 0.

Step 2: Accept the input value of n.

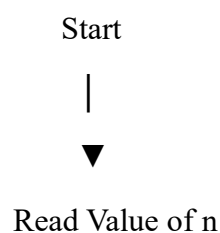
Step 3: Check whether n is greater than 0.

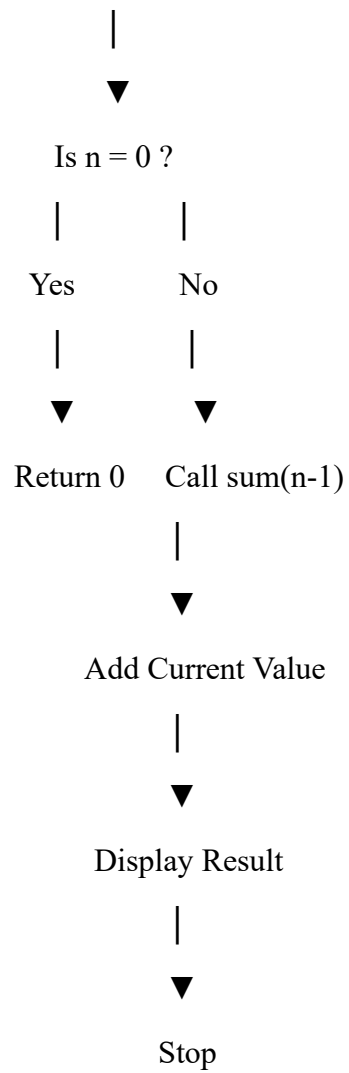
Step 4: Recursively calculate the sum of (n-1).

Step 5: Add n to the previous sum.

Step 6: Display the final sum.

Flowchart





Prolog Program

sum(0,0).

sum(N,S):-

N>0,

N1 is N-1,

sum(N1,S1),

S is S1+N.

Sample Output

?- sum(5,S).

S = 15.

?- sum(10,S).

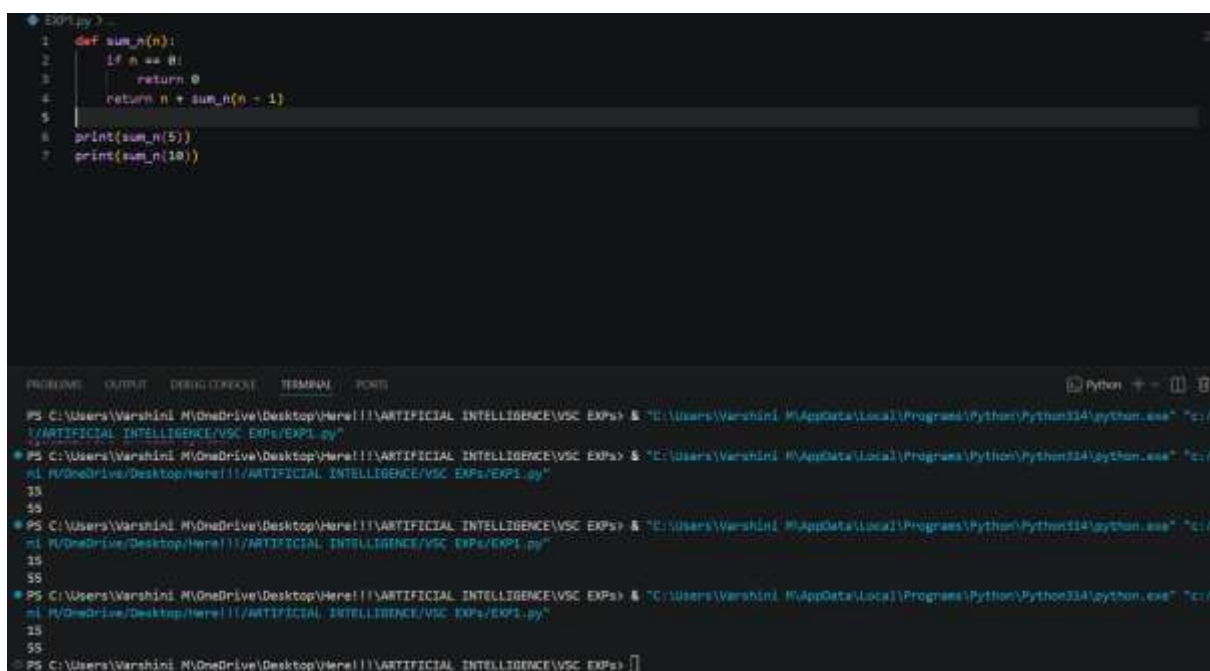
S = 55.

Result

The Prolog program successfully calculated the sum of integers from **1 to n** using recursion.

Conclusion

This experiment demonstrates the use of recursion in Prolog. Recursive predicates simplify mathematical computations and are widely used in Artificial Intelligence applications.



```
1 def sum_n(n):
2     if n == 0:
3         return 0
4     return n + sum_n(n - 1)
5
6 print(sum_n(5))
7 print(sum_n(10))
```

Python + - [Icons]

```
PS C:\Users\Varshini M\OneDrive\Desktop\Here\ARTIFICIAL INTELLIGENCE\VSC EXPS> "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\python.exe" "c:/i/ARTIFICIAL INTELLIGENCE/VSC EXPS/EXPI.py"
15
55
PS C:\Users\Varshini M\OneDrive\Desktop\Here\ARTIFICIAL INTELLIGENCE\VSC EXPS> "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\python.exe" "c:/i/ARTIFICIAL INTELLIGENCE/VSC EXPS/EXPI.py"
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PS C:\Users\Varshini M\OneDrive\Desktop\Here\ARTIFICIAL INTELLIGENCE\VSC EXPS> "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\python.exe" "c:/i/ARTIFICIAL INTELLIGENCE/VSC EXPS/EXPI.py"
15
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PS C:\Users\Varshini M\OneDrive\Desktop\Here\ARTIFICIAL INTELLIGENCE\VSC EXPS>
```